

# Advanced Topics in Computer Graphics I

## Filtering



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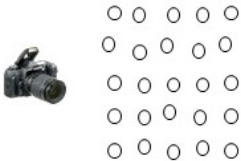
June 4, 2025



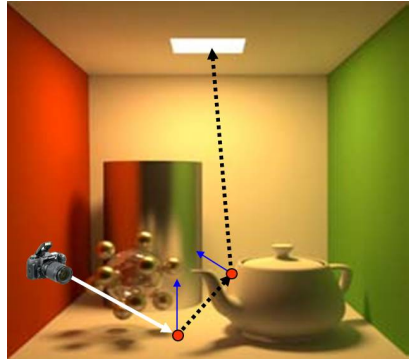


Monte-Carlo-Integration of the Rendering Equation was realized by the Path Tracing algorithm.

For each pixel, we shoot  $k$  primary rays.



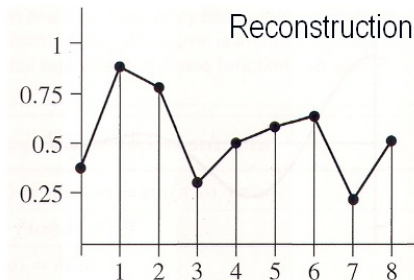
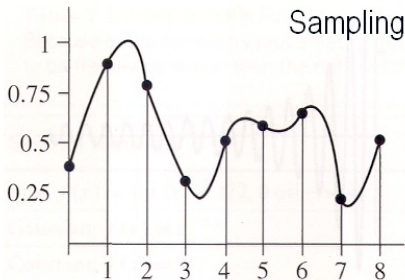
- ▶  $k = 1$  Directly use the computed color
- ▶  $k > 1$  For more than one color, we need a reconstruction filter!





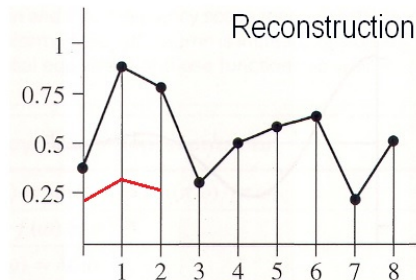
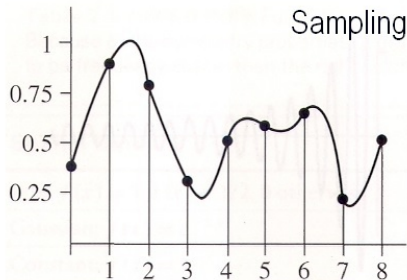
## Sampling and Reconstruction





**Question:** What is meant by sampling and reconstruction?

- ▶ **Sampling:** (Discrete) sampling of an unknown function
- ▶ **Reconstruction:** Reconstruction of the function from samples



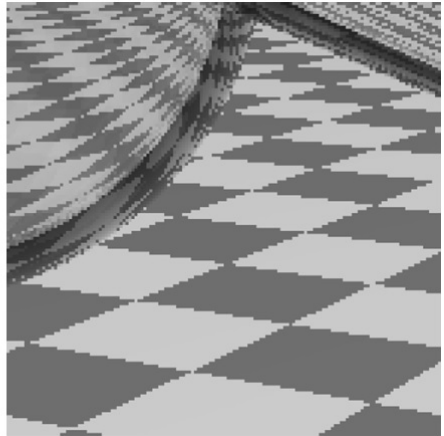
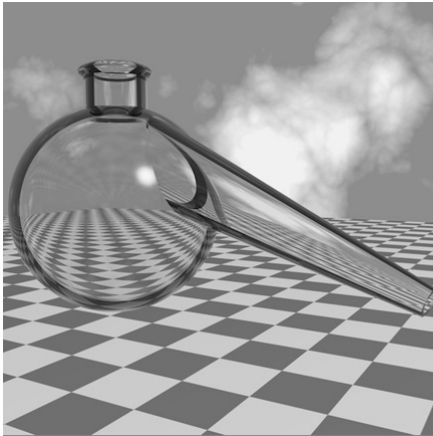
**Problem:** Reconstruction only approximates the original function

→ Reconstruction error

→ Aliasing



## Example: “Jaggies”

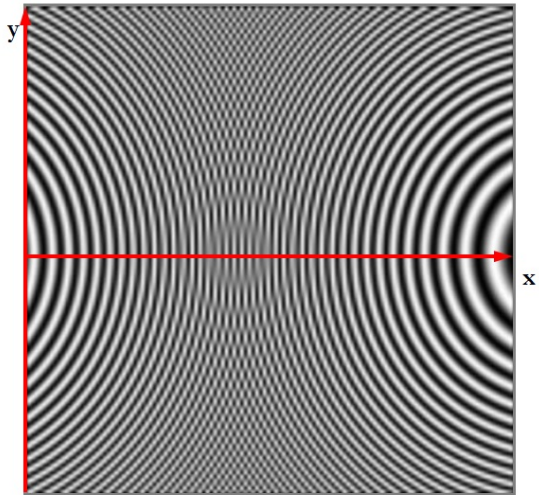




## Example: Moiré Pattern

Sampling of the function:

$$\sin(x^2 + y^2)$$





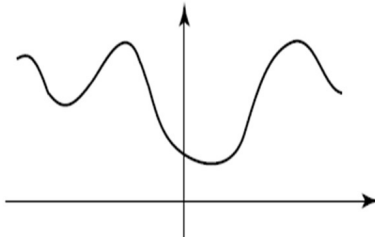
**Question:** What happens during sampling and reconstruction?

**Idea:** Fourier Analysis

Given a continuous function (signal)  $f$ :

$$f: \mathbb{R} \rightarrow \mathbb{C}.$$

We interpret this function as a point in an infinite-dimensional vector space. The  $i$ -th value of  $f$  describes the distance of the function to the  $i$ -th dimension of this vector space  $V$ .







Since our functions are now interpreted as points in a vector space  $V$ , we can define a dot product and a norm:

## Definition (Dot product between signals)

Let  $f, g: \mathbb{R} \rightarrow \mathbb{C}$  be continuous functions. The *dot product* between  $f$  and  $g$  is defined as

$$\langle f | g \rangle_V := \int_{\mathbb{R}} f(t) \overline{g(t)} dt$$

## Definition (Norm of a signal)

Let  $f: \mathbb{R} \rightarrow \mathbb{C}$  be a continuous function. The *2-norm* of  $f$  is defined as

$$\|f\|_2 := \sqrt{\langle f | f \rangle_V}$$

Furthermore, the following holds:

$$\|f\|_2 = \sqrt{\int_{\mathbb{R}} |f(t)|^2 dt}$$



To analyze a function, we need one important property:

Definition (Space of signals with finite energy)

The set

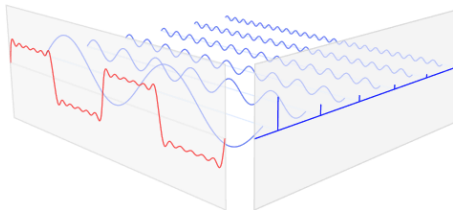
$$L^2(\mathbb{R}) := \{f: \mathbb{R} \rightarrow \mathbb{C} \mid \|f\|_2 < \infty\}$$

is called the *Space of signals with finite energy*.

Typically, all functions which we will consider lie in this space. So, we do not restrict ourselves with this property.

## Crucial observation:

Any signal can be decomposed into  
an infinite sum of sine and cosine  
functions!





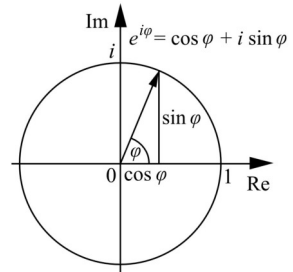
Using this observation, we derive the Fourier Transform:

Lemma (Euler's formula)

*The equation*

$$e^{ix} = \cos(x) + i \sin(x)$$

*is called Euler's formula.*



So we can write the complex sum of a sine and cosine as an exponential of a complex number. Furthermore, we can use the fact that the functions

$$e_{\omega}: t \mapsto e^{2\pi i \omega t}$$

span an **orthonormal basis** over the so called **frequency domain**. Since this basis is orthonormal, we can compute the component of  $f$  to each basis function using the dot product.



Since this basis is orthonormal, we can compute the component of  $f$  to each basis function using the dot product.

## Definition (Fourier Transform)

Let  $f \in L^2(\mathbb{R})$ . The function  $\hat{f}: \mathbb{R} \rightarrow \mathbb{C}$  with

$$\hat{f}(\omega) = \int_{\mathbb{R}} f(t) e^{-2\pi i \omega t} dt$$

is called *Fourier Transform* of  $f$ .

On the other hand, we can write the signal in this basis. This is called Inverse Fourier Transform:

## Definition (Inverse Fourier Transform)

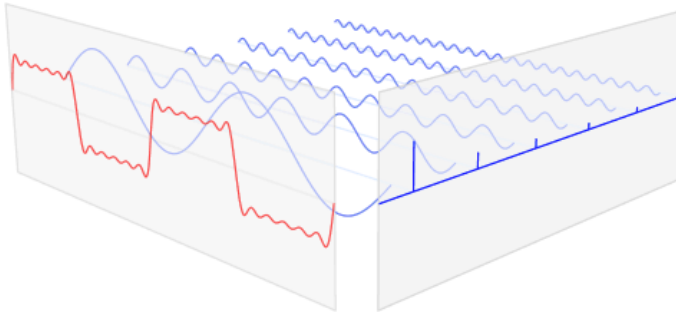
Let  $\hat{f} \in L^2(\mathbb{R})$ . The mapping  $f: \mathbb{R} \rightarrow \mathbb{C}$  with

$$f(t) = \int_{\mathbb{R}} \hat{f}(\omega) e^{2\pi i \omega t} d\omega$$

is called *Inverse Fourier Transform* of  $\hat{f}$ .



So, we discovered the fundamental formulae to convert a signal between spatial and frequency domain:



Spatial domain

$$f(t) = \int_{\mathbb{R}} \hat{f}(\omega) e^{2\pi i \omega t} d\omega$$

Frequency domain

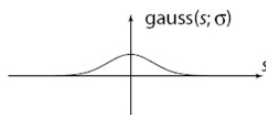
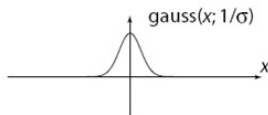
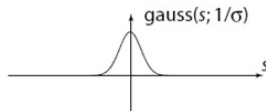
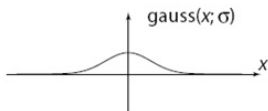
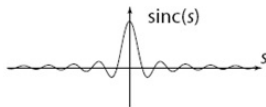
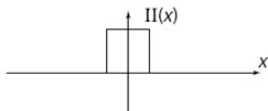
$$\hat{f}(\omega) = \int_{\mathbb{R}} f(t) e^{-2\pi i \omega t} dt$$



Example:

spatial domain

frequency domain





## Definition (2D Fourier Transform)

Let  $f \in L^2(\mathbb{R}^2)$ . The function  $\hat{f}: \mathbb{R}^2 \rightarrow \mathbb{C}$  with

$$\hat{f}(u, v) = \int_{\mathbb{R}} \int_{\mathbb{R}} f(x, y) e^{-2\pi i(ux+vy)} dx dy$$

is called *Fourier Transform* of  $f$ , where  $u$  and  $v$  are spatial frequencies.  
The corresponding mapping

$$f(x, y) = \int_{\mathbb{R}} \int_{\mathbb{R}} \hat{f}(u, v) e^{2\pi i(ux+vy)} du dv$$

is called *Inverse Fourier Transform* of  $\hat{f}$ .

## Remarks

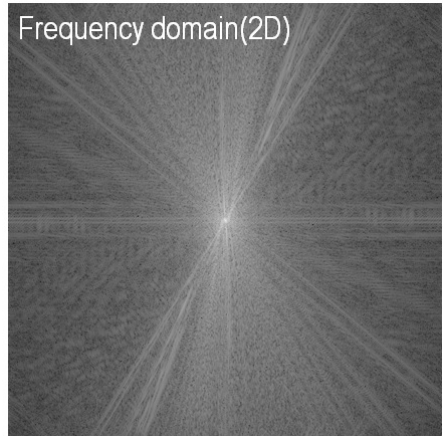
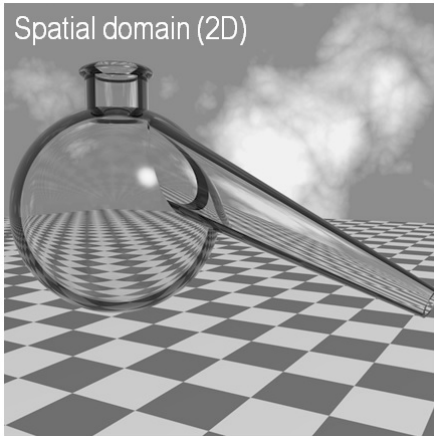
- ▶  $\hat{f}(u, v)$  is complex in general,

$$\hat{f}(u, v) = \hat{f}_{\text{Re}}(u, v) + i\hat{f}_{\text{Im}}(u, v)$$

- ▶  $|\hat{f}(u, v)|$  is called the magnitude spectrum
- ▶  $\arctan\left(\hat{f}_{\text{Im}}(u, v)/\hat{f}_{\text{Re}}(u, v)\right)$  is called the phase angle spectrum



**Example:**

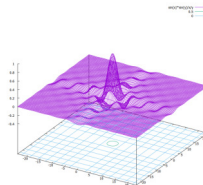
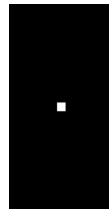






**Example 1:** rectangle of height 1 centered at origin with sides of length  $X$  and  $Y$

$$\begin{aligned}\hat{f}(u, v) &= \int_{\mathbb{R}} \int_{\mathbb{R}} f(x, y) e^{-2\pi i(ux+vy)} dx dy \\&= 1 \cdot \int_{-\frac{X}{2}}^{\frac{X}{2}} e^{-2\pi i u x} dx \cdot \int_{-\frac{Y}{2}}^{\frac{Y}{2}} e^{-2\pi i v y} dy \\&= \left[ \frac{e^{-2\pi i u x}}{-i2\pi u} \right]_{-\frac{X}{2}}^{\frac{X}{2}} \cdot \left[ \frac{e^{-2\pi i v y}}{-i2\pi v} \right]_{-\frac{Y}{2}}^{\frac{Y}{2}} \\&= \frac{1}{-2\pi i u} [e^{-i\pi u X} - e^{i\pi u X}] \cdot \frac{1}{-2\pi i v} [e^{-i\pi v Y} - e^{i\pi v Y}] \\&= XY \left[ \frac{\sin(\pi X u)}{\pi X u} \right] \left[ \frac{\sin(\pi Y v)}{\pi Y v} \right] \\&= XY \cdot \text{sinc}(\pi X u) \cdot \text{sinc}(\pi Y v).\end{aligned}$$





In order to understand filtering, we need one more operator:

## Definition (Convolution)

Let  $f, g \in L^2(\mathbb{R})$  be two signals. The *convolution* between  $f$  and  $g$  is defined as

$$(f * g)(t) := \int_{\mathbb{R}} f(t - x) g(x) dx$$

The convolution of two signals is again a signal.

The convolution has the following properties:

## Theorem (Algebraic properties of the convolution)

Let  $f, g, h \in L^2(\mathbb{R})$  be three signals and  $\alpha, \beta \in \mathbb{C}$  be complex numbers. The convolution is associative, commutative and linear:

- ▶  $(f * g) * h = f * (g * h)$
- ▶  $f * g = g * f$
- ▶  $f * (\alpha g + \beta h) = \alpha (f * g) + \beta (f * h)$

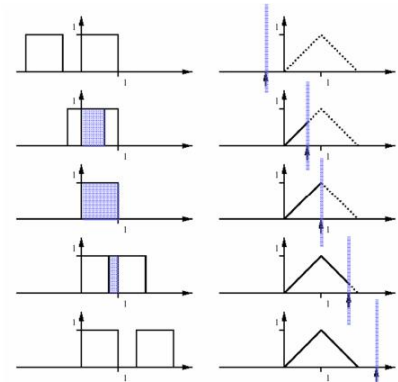


# Convolution



Intuitively, the convolution between two signals  $f$  and  $g$  is the intersected area between the **mirrored and shifted signal  $f$**  and the **signal  $g$** .

$$(f * g)(t) := \int_{\mathbb{R}} f(t - x) g(x) dx$$





# Convolution

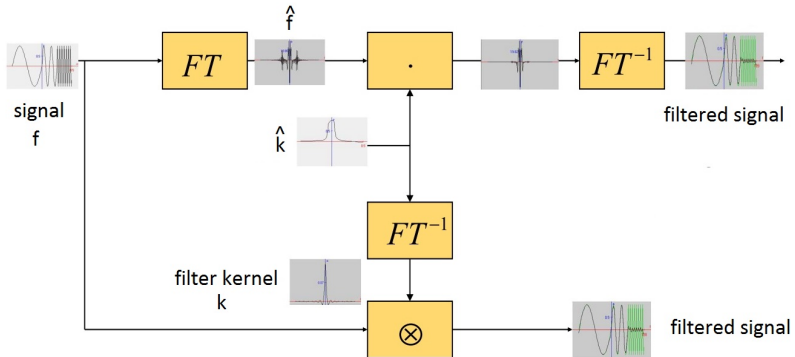


Now, we can formulate the Convolution Theorem:

## Theorem (Convolution Theorem)

Let  $f, g \in L^2(\mathbb{R})$  be two signals. Then the following holds:

$$\widehat{f * g} = \widehat{f} \cdot \widehat{g}$$



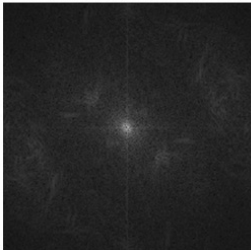
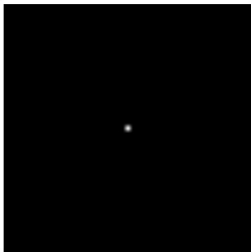
Convolution in spatial domain is equal to point wise multiplication in frequency domain.



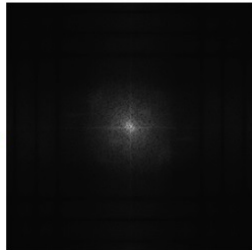
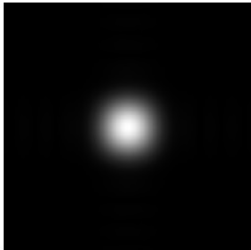
Example:



\*



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# Sampling Theorem

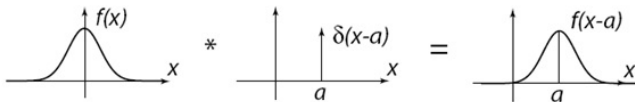


Now, we will derive the Sampling Theorem which will give us the reason why aliasing occurs during sampling.

Consider the Dirac Delta function. If we convolve a function  $f$  with  $\delta$ , we get the function value of  $f$ :

$$\begin{aligned}(f * \delta)(x_0) &= \int_{\mathbb{R}} f(x_0 - x) \delta(x) dx \\ &= \int_{\mathbb{R}} \delta(x_0 - x) f(x) dx \\ &= f(x_0)\end{aligned}$$

Similarly, if we convolve a shifted Dirac Delta function with  $f$ , we get a shifted  $f$ :





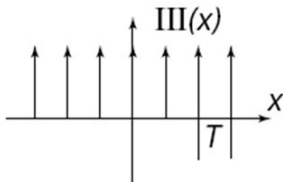
# Sampling Theorem



Now, we consider the Shah-function  $\text{III}(x)$  which is a sum of shifted Delta functions:

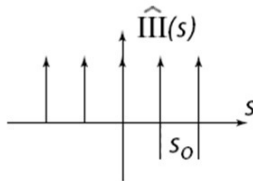
Spatial domain

$$\text{III}(x) = \sum_{n=-\infty}^{\infty} \delta(x - nT)$$



Frequency domain

$$\hat{\text{III}}(s) = \sum_{n=-\infty}^{\infty} \delta(s - ns_0) \quad s_0 = 1/T$$



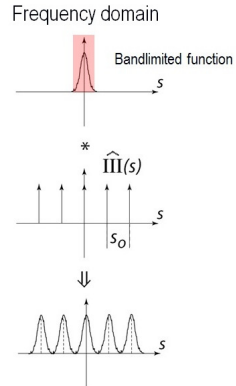
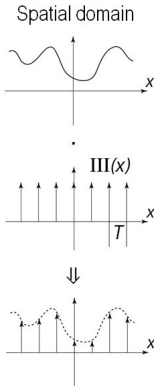
**Observation:** The Shah-function transformed to the frequency domain is again a Shah-function!



# Sampling Theorem



Sampling can be regarded as point-wise multiplication of the Shah-function to the continuous signal  $f$ :



**Observation:** In frequency domain, this leads to copies of  $\hat{f}$ !

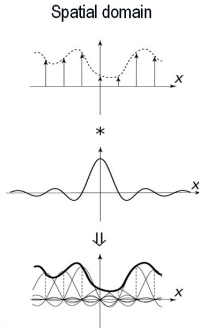




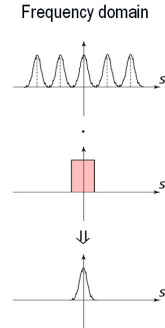
# Sampling Theorem



To reconstruct the original signal, we need to point-wise multiply the signal in frequency domain with a box filter. In spatial domain, this is a convolution with the sinc-function:



$$\begin{aligned}\tilde{f} &= (f(x) \cdot \text{III}(x)) * \text{sinc}(x) \\ &= \sum_{n=-\infty}^{\infty} \text{sinc}(x - n) f(n)\end{aligned}$$



$$\tilde{f} = (\hat{f}(s) * \hat{\text{III}}(s)) \cdot \Pi(s)$$



# Sampling Theorem

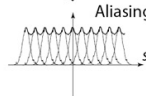
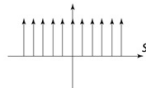
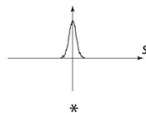
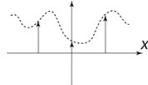
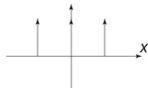
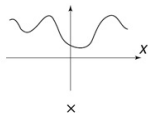


Aliasing occurs if the sampling of the function was not dense enough. Then the copies of  $\hat{f}$  in the frequency domain **overlap** and cause artifacts during reconstruction:

Large sample distance  
in spatial domain



Small distance in  
frequency domain



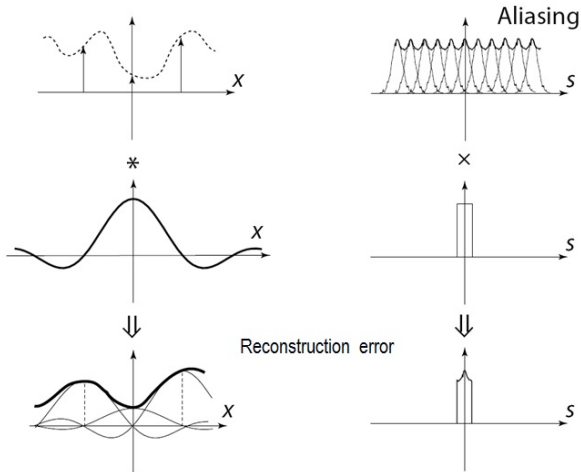
High frequency details overlap  
with low frequency ones



# Sampling Theorem



Aliasing occurs if the sampling of the function was not dense enough. Then the copies of  $\hat{f}$  in frequency domain **overlap** and cause artifacts during reconstruction:





## Theorem (Sampling Theorem<sup>1</sup>)

*If the function  $f \in L^2(\mathbb{R})$  is band-limited, i.e.*

$$\exists \omega_{\max} : \forall \omega > \omega_{\max} : \hat{f}(\omega) = 0,$$

*then the function  $f$  can be reconstructed from the the samples  $f(m \cdot \Delta t)$  without any error using a sampling frequency of at least*

$$\frac{1}{\Delta t} \geq 2 \cdot \omega_{\max}.$$

*This sampling frequency is called Nyquist Frequency.*

**Anti-Aliasing-Techniques:** To avoid aliasing, there are two major techniques:

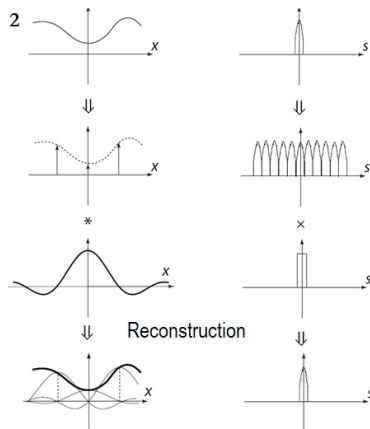
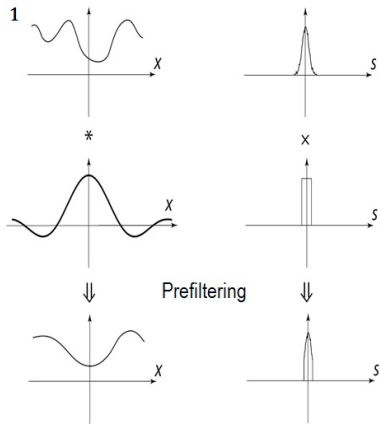
- ▶ **Prefiltering:** Band-limit the signal via filtering
- ▶ **Supersampling:** Increase the sampling rate

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<sup>1</sup>C. E. Shannon. "Communication in the presence of noise". In: *Proceedings of the IRE* 37.1 (1949), pp. 10–21.



## Technique 1: Prefiltering



*"blurring is better than aliasing"*

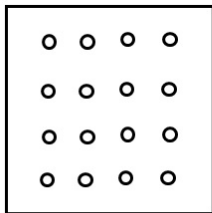


- ▶ **Natural Perception:** Blurring mimics real-world defocusing, making it less jarring and more interpretable to the human eye compared to the artificial patterns of aliasing.
- ▶ **Information Preservation:** Blurring maintains underlying content in a smoothed form, while aliasing introduces false information, which can be misleading especially in critical applications like medical imaging.
- ▶ **Visual Appeal:** Blurring is subtler and often more aesthetically pleasing, whereas aliasing creates harsh and distracting visual artifacts.
- ▶ **Post-Processing Potential:** Blurring allows for the possibility of information recovery through deblurring techniques, while the information distorted by aliasing is generally irretrievable.
- ▶ **Predictable Error:** Blurring produces consistent and uniform errors, in contrast to the unpredictable and variable high-frequency artifacts caused by aliasing.



## Technique 2: Supersampling

- **Assumption:** Energy (Fourier Transform) of a function decreases with increasing frequency
- **Supersampling:** Increase of sampling rate  $\rightarrow$  Reduction of “aliasing energy”
- **Resampling:** Low-pass filter to obtain desired discretization

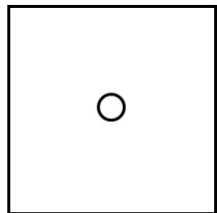


**Samples**

$$Pixel = \sum_s w_s \cdot Sample_s$$



Resampling



**Pixel**



**Example:**

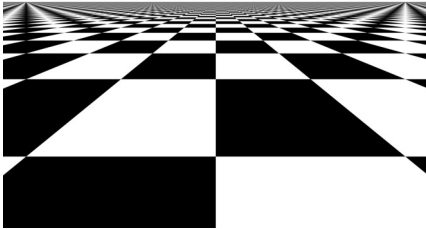
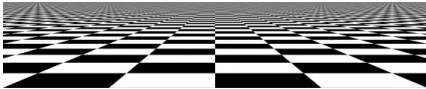


Figure: Exact (analytic)

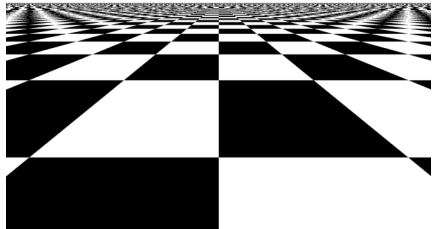


Figure: Point sampling





**Example:**

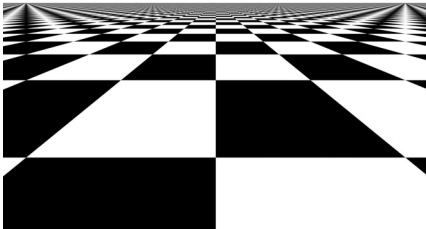


Figure: Exact (analytic)

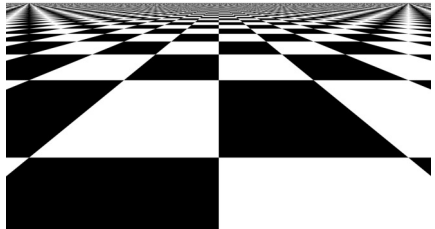
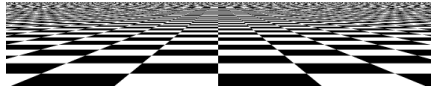


Figure: 4x4 Regular super sampling



**Example:**

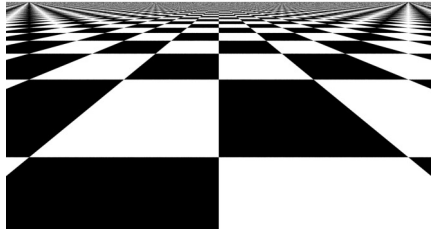
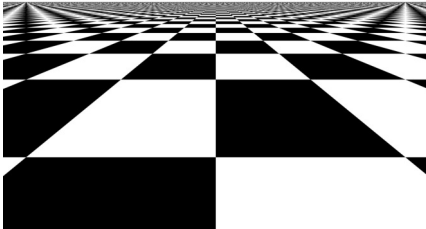
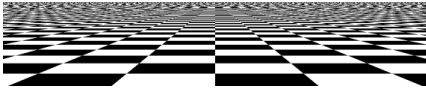


Figure: 4x4 Regular super sampling

Figure: 4x4 Jittered super sampling

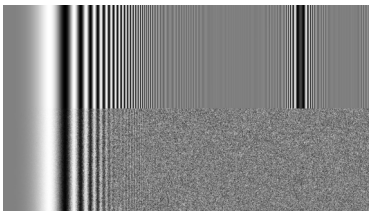


## ► Uniform sampling

- The spectrum of uniformly spaced samples is also a set of uniformly spaced spikes
- Multiplying the signal by the sampling pattern corresponds to placing a copy of the spectrum at each spike (in freq.space) resulting in coherent (structured) aliasing
- Structures are noticeable!

## ► Non-uniform sampling

- Samples at non-uniform locations have a different spectrum; a single spike plus noise
- Sampling a signal in this way converts aliases into broadband noise
- Noise is incoherent (structureless) and much less objectionable



- The impulse train is modified as

$$\sum_{i=-\infty}^{\infty} \delta\left(x - \left(i + \frac{1}{2} - \xi\right)T\right),$$

where  $\xi \in [0, 1]$  is a random number.

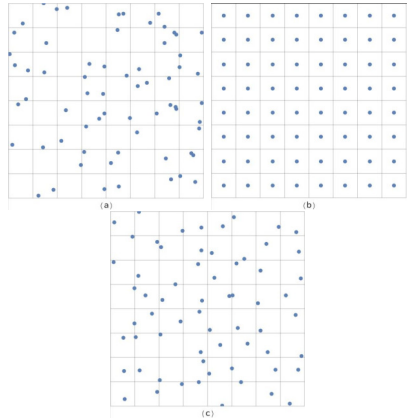
- It turns regular aliasing artifacts into noise, which is less distracting to the human visual system.



# Stratified sampling



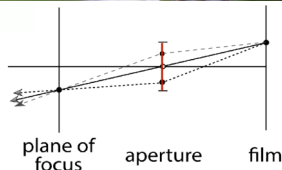
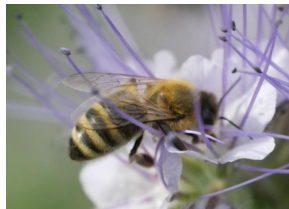
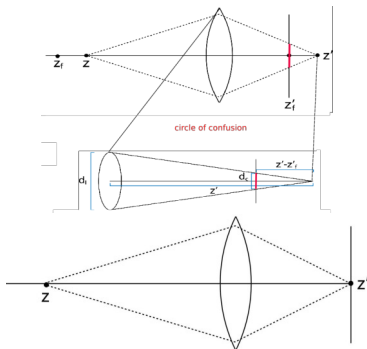
- ▶ The **random pattern** is an ineffective pattern, with many clumps of samples that leave large sections of the image poorly sampled.
- ▶ A **uniform stratified pattern** is better distributed but can exacerbate aliasing artifacts
- ▶ A **stratified jittered pattern** turns aliasing from the uniform pattern into high-frequency noise while still maintaining the benefits of stratification





**Curse of dimensionality** Direct application of stratification to high-dimensional sampling quickly leads to an intractable number of samples.

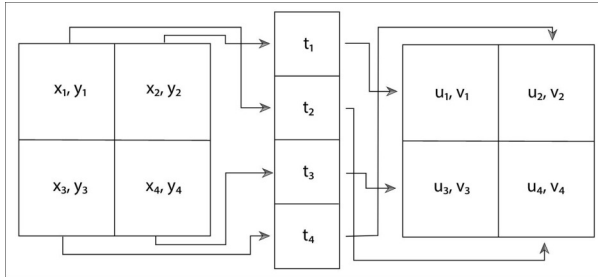
**Example** (depth of field, motion blur)



If we divided the 5D image, lens, and time sample space into four strata in each dimension, the total number of samples per pixel would be 1024.



## Padding (to counter curse of dimensionality )

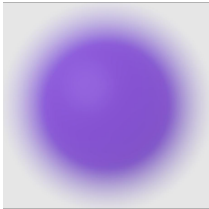


- ▶ Image position, time, and lens position are split into independent strata with four regions each.
- ▶ Each strata is sampled independently.
- ▶ A time sample and a lens sample are randomly associated with each image sample.

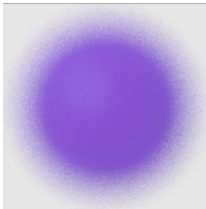
We retain the benefits of stratification in each of the individual dimensions without having to exponentially increase the total number of samples.



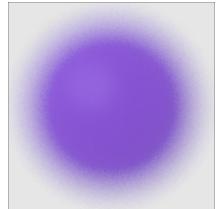
## Example



a) Reference image



b) Random sampling

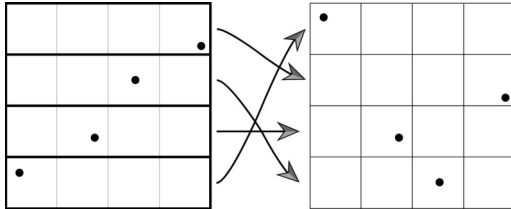


c) Stratified sampling with same number of samples.





**Challenge** How to distribute an arbitrary number of samples in  $[0, 1)^s$ ?

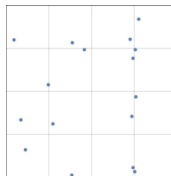


## Algorithm

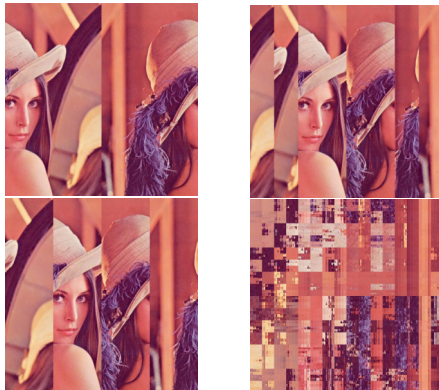
- ▶ Divide each dimension's axes into  $n$  regions and generate jittered sample along the diagonal
- ▶ Shuffle the samples randomly in each dimension

**Remark** In two dimensions a square grid containing sample positions is a Latin square if (and only if) there is only one sample in each row and each column. Generalized to higher dimensions each sample is the only one in each axis-aligned hyperplane.

- ▶ LHS avoids clumping of the samples when they are projected onto any of the axes of the sampling dimensions like this might happen for stratified sampling as shown right.



- ▶ Shuffling does not influence stratification properties





To evaluate the quality of a pattern of  $n$ -dimensional sample positions, especially if a set of random points covers a certain area/volume equally dense the concept of discrepancy was introduced<sup>2</sup>.

## Definition (Discrepancy, low-discrepancy sequence)

The discrepancy of a point set  $P$  with respect to a family of shapes  $B$  that are subsets of  $[0, 1)^d$  is defined as

$$D_N(P, B) = \sup_{b \in B} \left| \frac{|\{x_i \in b\}|}{N} - V(b) \right|,$$

where  $|\{x_i \in b\}|$  is the number of points in  $b$  and  $V(b)$  is the volume of  $b$ .

A **low-discrepancy sequence** is a sequence with the property that for all values of  $k$ , its subsequence  $x_1, \dots, x_k$  has a low discrepancy.

---

<sup>2</sup>H. Weyl. "Über die gleichverteilung von zahlen mod. eins". In: *Mathematische Annalen* 77.3 (1916), pp. 313–352.



Note, that the discrepancy of a sequence is low if the proportion of points in the sequence falling into an arbitrary set  $b$  is close to proportional to the volume of  $b$ , as would happen on average (but not for particular samples) in the case of an equi-distributed sequence.

## Example

Let  $P = \{x_1, x_2, \dots\}$  a sequence of points in  $[a, b] \subset \mathbb{R}$ .

We choose  $B = \{[c, d] \subset [a, b]\}$ . Then the discrepancy is given by

$$D_N(P, B) = \sup_{a \leq c \leq d \leq b} \left| \frac{|\{x_1, \dots, x_n\} \cap [c, d]|}{N} - \frac{d - c}{b - a} \right|,$$



## Definition (Star Discrepancy)

If the set of shapes  $B$  is the set of boxes with a corner at the origin, the discrepancy is called **star discrepancy**  $D_n^*(P)$ .

For a few particular point sets, the discrepancy can be computed analytically. If  $x_i = \frac{i}{N}, i = 1, \dots, n$  we can see that the star discrepancy is

$$D_N^*(x_1, \dots, x_n) = \frac{1}{N}.$$

The star discrepancy of this sequence can be improved by modifying it slightly:

$$x_i = \frac{i - 0.5}{N}, i = 1, \dots, n$$

Then

$$D_N^*(x_1, \dots, x_n) = \frac{1}{2N}.$$

**Bounds of the star discrepancy** It can be shown that in one dimension we get

$$D_N^*(x_1, \dots, x_n) \leq \frac{1}{2N} + \max_{1 \leq i \leq N} \left| x_i - \frac{2i - 1}{2N} \right|.$$



## Uniform sampling techniques<sup>3</sup>: Radical Inversion

- Van der Corput sequence in base  $b$

$$\Phi_b : \mathbb{N}_0 \rightarrow \mathbb{Q} \cap [0, 1)$$

$$i = \sum_{l=0}^{\infty} a_l(i) b^l \mapsto \Phi_b(i) := \sum_{l=0}^{\infty} a_l(i) b^{-l-1}.$$



| $a$ | Base 2 | $\Phi_2(a)$   |
|-----|--------|---------------|
| 0   | 0      | 0             |
| 1   | 1      | $0.1 = 1/2$   |
| 2   | 10     | $0.01 = 1/4$  |
| 3   | 11     | $0.11 = 3/4$  |
| 4   | 100    | $0.001 = 1/8$ |
| 5   | 101    | $0.101 = 5/8$ |

- Matrix representation (finite sequence)

$$\Phi_b(i) = [b^{-1}b^{-2} \dots b^{-n}] \begin{bmatrix} a_0(i) \\ a_1(i) \\ \vdots \\ a_{n-1}(i) \end{bmatrix} = [b^{-1}b^{-2} \dots b^{-n}] \textcolor{green}{I} \begin{bmatrix} a_0(i) \\ a_1(i) \\ \vdots \\ a_{n-1}(i) \end{bmatrix}$$

<sup>3</sup>A. Keller, I. Georgiev, et al. "My favorite samples". In: *ACM SIGGRAPH 2019 Courses*. ACM, 2019, pp. 1–271.



## Properties: Radical Inversion

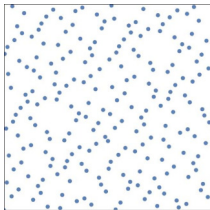
- ▶ subsequent points that “fall into biggest holes”
- ▶ not completely uniform distributed (CUD)
- ▶ contiguous blocks of stratified points  $x_i$  for  $kb^m \leq i < (k+1)b^m - 1$ 
  - ▶ for each block the  $\Phi_b(i)$  are equidistant
  - ▶ for each block the integers  $\lfloor b^m \Phi_b(i) \rfloor$  are a **permutation** of  $\{0, \dots, b^m - 1\}$
- ▶ The discrepancy of this sequence is

$$D_N^*(\Phi_b) = \mathcal{O}\left(\frac{\log N}{N}\right).$$

- Let the  $b_j$  be co-prime, for example the  $j$ -th prime number

Halton sequence

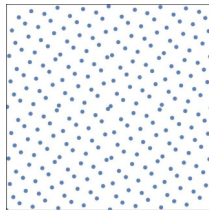
$$x_i := (\Phi_{b_1}(i), \dots, \Phi_{b_s}(i))$$



$$(\Phi_2(i), \Phi_3(i))_{i=0}^{215}$$

Hammersley sequence

$$x_i := \left( \frac{i}{n}, \Phi_{b_1}(i), \dots, \Phi_{b_{s-1}}(i) \right)$$



$$\left( \frac{i}{256}, \Phi_2(i) \right)_{i=0}^{255}$$

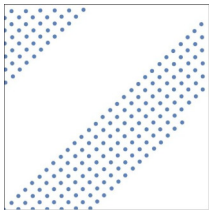
- Contiguous blocks of stratified points  $x_i$  for  $k \prod_{j=1}^s b_j^{m_j} \leq i < (k+1) \prod_{j=1}^s b_j^{m_j} - 1$



- Let the  $b_j$  be co-prime, for example the  $j$ -th prime number

Halton sequence

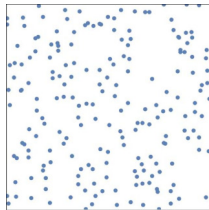
$$x_i := (\Phi_{b_1}(i), \dots, \Phi_{b_s}(i))$$



$$(\Phi_{29}(i), \Phi_{31}(i))_{i=0}^{215}$$

Halton scrambled

$$x_i := (\Psi_{b_1}(i), \dots, \Psi_{b_s}(i))$$



$$(\Psi_{29}(i), \Psi_{31}(i))_{i=0}^{215}$$

- Correlations in low dimensional projections can be avoided by applying a permutation to the digits when computing the radical inverse:

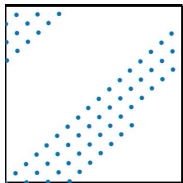
$$i = \sum_{l=0}^{\infty} a_l(i) b^l \mapsto \Psi_b(i) := \sum_{l=0}^{\infty} \sigma(a_l(i)) b^{-l-1}.$$

## Scrambled radical inversion

- ▶ example: deterministic permutations  $\sigma_b$  by Faure

$$i = \sum_{l=0}^{\infty} a_l(i) b^l \mapsto \Psi_b(i) := \sum_{l=0}^{\infty} \sigma_b(a_l(i)) b^{-l-1}.$$

- ▶  $b$  is even: Take  $2\sigma_{\frac{b}{2}}$  and append  $2\sigma_{\frac{b}{2}} + 1$
- ▶  $b$  is odd: Take  $\sigma_{b-1}$ , increment each value  $\geq \frac{b-1}{2}$  and insert  $\frac{b-1}{2}$  in the middle

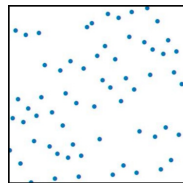


$(\Phi_{17}(i), \Phi_{19}(i))_{i=0}^{63}$

→

$$\begin{aligned}\sigma_2 &= (0, 1) \\ \sigma_3 &= (0, 1, 2) \\ \sigma_4 &= (0, 2, 1, 3) \\ \sigma_5 &= (0, 3, 2, 1, 4) \\ \sigma_6 &= (0, 2, 4, 1, 3, 5)\end{aligned}$$

→



$(\Psi_{17}(i), \Psi_{19}(i))_{i=0}^{63}$

## Definition

## ► Elementary interval

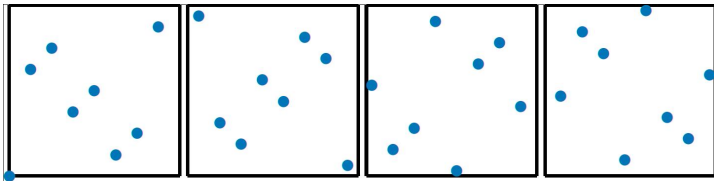
$$E := \prod_{j=1}^s \left[ \frac{a_j}{b^{I_j}}, \frac{a_j + 1}{b^{I_j}} \right) \subset I^s \text{ for integers } I_j \geq 0 \text{ and } 0 \leq a_j < b^{I_j}$$

with volume  $\lambda_s(E) = \prod_{j=1}^s \frac{1}{b^{I_j}} = b^{-\left(\sum_{j=1}^s I_j\right)}$

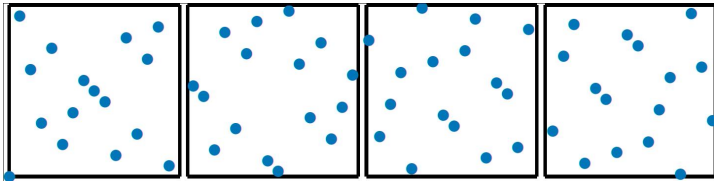
- For two integers  $0 \leq t \leq m$ , a finite point set of  $b^m$  points in  $s$  dimensions is called a  $(t, m, s)$ -net in base  $b$ , if every elementary interval of volume  $\lambda_s(E) = b^{t-m}$  contains exactly  $b^t$  points.
- For  $t \geq 0$ , an infinite point sequence is called a  $(t, s)$ -sequence in base  $b$ , if for all  $k \geq 0$  and  $m \geq t$ , the vectors  $x_{kb^m}, \dots, x_{(k+1)b^m-1} \in I^s$  form a  $(t, m, s)$ -net.

**Example** stratification properties of the Sobol'  $(0, 2)$ -sequence<sup>4</sup> in base 2

- The sequence of  $(0, 3, 2)$ -nets



- The sequence of  $(0, 4, 2)$ -nets

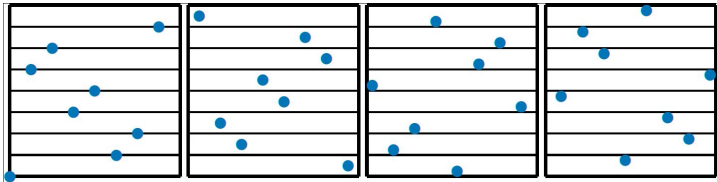


<sup>4</sup>A. Keller, S. Premoze, et al. "Advanced (quasi) Monte Carlo methods for image synthesis". In: *ACM SIGGRAPH 2012 Courses*. 2012, pp. 1–46.

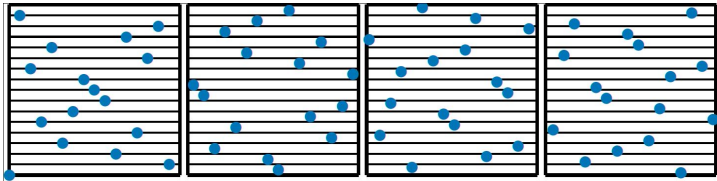


## Example stratification properties of the Sobol' $(0, 2)$ -sequence in base 2

- The sequence of  $(0, 3, 2)$ -nets



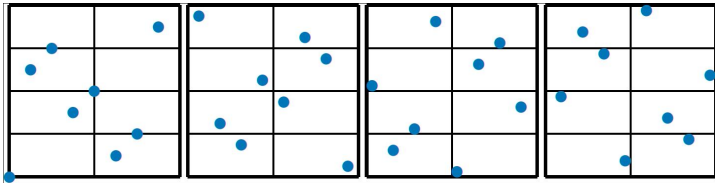
- The sequence of  $(0, 4, 2)$ -nets



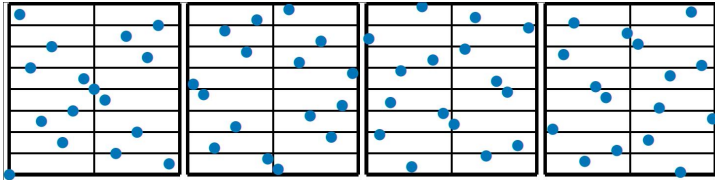


## Example stratification properties of the Sobol' $(0, 2)$ -sequence in base 2

- The sequence of  $(0, 3, 2)$ -nets

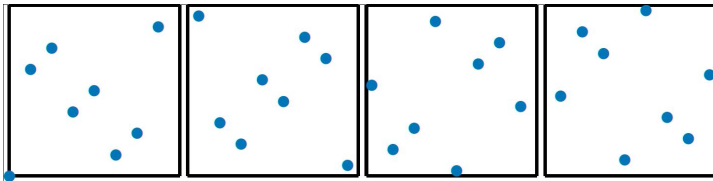


- The sequence of  $(0, 4, 2)$ -nets

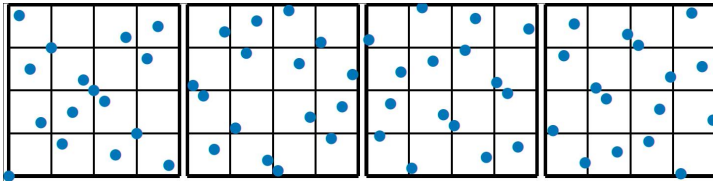


**Example** stratification properties of the Sobol'  $(0, 2)$ -sequence in base 2

- The sequence of  $(0, 3, 2)$ -nets



- The sequence of  $(0, 4, 2)$ -nets



- All components of the Sobol' sequence are  $(0, 1)$ -sequences in base 2  $\Rightarrow$  deterministic LHS

## Generator matrices as generalization of radical inversion<sup>5,6</sup>

- Use one  $m \times m$  generator matrix  $C^j$  for each component

$$x_i^{(j)} = \Phi_b^j(i) \underbrace{\left[ b^{-1}b^{-2} \dots b^{-n} \right] C^j}_{\text{multiplication in } \mathbb{F}_b} \begin{bmatrix} a_0(i) \\ a_1(i) \\ \vdots \\ a_{n-1}(i) \end{bmatrix}$$

- This multiplication can be performed very efficiently e.g. in  $\mathbb{Z}_2$ , since in this setting addition corresponds to the **exclusive OR** operation.

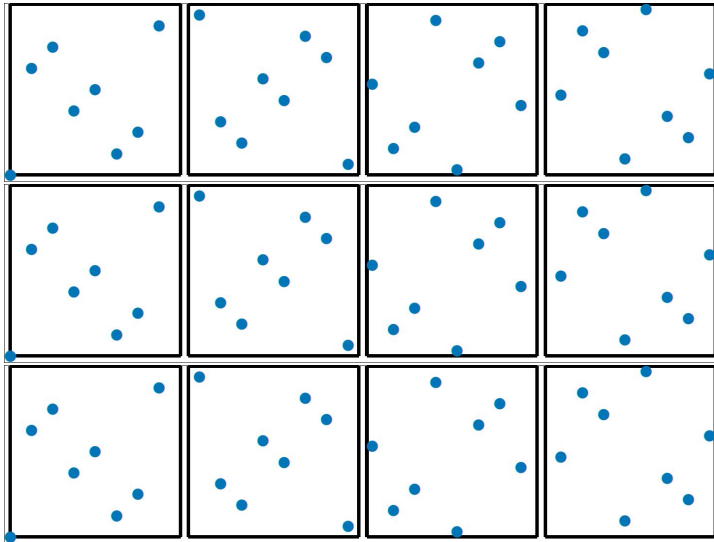
$$C^j \begin{bmatrix} a_0(i) \\ a_1(i) \\ \vdots \\ a_{n-1}(i) \end{bmatrix} = a_0(i) \begin{bmatrix} c_{11} \\ c_{21} \\ \vdots \\ c_{n1} \end{bmatrix} + a_1(i) \begin{bmatrix} c_{12} \\ c_{22} \\ \vdots \\ c_{n2} \end{bmatrix} + \dots + a_{n-1}(i) \begin{bmatrix} c_{1n} \\ c_{2n} \\ \vdots \\ c_{nn} \end{bmatrix}$$

<sup>5</sup>S. Joe and F. Y. Kuo. *Sobol sequence generator*. 2020 (accessed May, 23, 2020). URL: <https://web.maths.unsw.edu.au/~fkuo/sobol/index.html>.

<sup>6</sup>L. Gruenschloss. *Quasi Monte Carlo Methods*. 2020 (accessed May, 23, 2020). URL: [gruenschloss.org](http://gruenschloss.org).



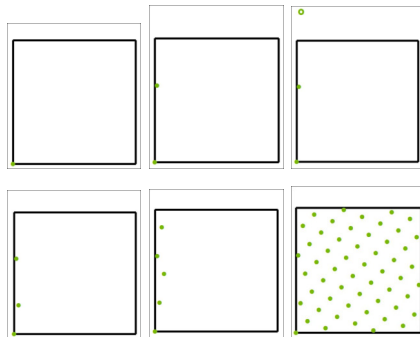
# Digital $(t, s)$ -sequences in base $b$





- ▶ Given generator vector  $(g_0, \dots, g_{s-1}) \in \mathbb{N}^s$

$$x_i := \frac{i}{n}(g_0, \dots, g_{s-1}) \mod [0, 1)^s$$



- ▶ Generator vectors
  - ▶ Korobov form  $(1, a, a^2, a^3, \dots)$
- ▶ Rare constructions
  - ▶ Example: Fibonacci lattice with  $n = F_k$  and  $(g_0, g_1) = (1, F_{k-1})$
- ▶ Usually tabulated coefficients  $a$  of  $g_j$ 
  - ▶ Search by certain criteria, e.g. maximized minimum distance, projections, ...
  - ▶ Component by component construction (CBC)



The **quasi-Monte Carlo method** uses low-discrepancy sequences instead of sequences of pseudo-random numbers for the approximation of the integral of a function  $f$  as the average of the function evaluated at a set of points  $\{x_1, \dots, x_N\}$  :

$$\int_{[0,1]^s} f(x) dx \approx \frac{1}{N} \sum_{i=1}^N f(x_i)$$

The error of the approximation depends on two independent factors:

- ▶ the **discrepancy** of the sequence of sample points
- ▶ The **variation** of the function  $f$ , i.e. how rapidly the function  $f$  changes

## Definition (Variation)

Let  $f$  be a real-valued function on  $[0, 1]$  and  $\mathcal{P} = \{0 = x_0 < x_1 < \dots < x_N = 1\}$  the set of all partitions of  $[0, 1]$ . Then

$$\mathcal{V}_V(f) = \limsup_{\mathcal{P}} \sum_{i=1}^N |f(x_{i+1}) - f(x_i)|$$

is called the **variation in the sense of Vitali** of  $f$ .



## Definition (Variation in higher dimensions)

For 2-dimensional functions the variation in the sense of Vitali is defined analogous:

$$\mathcal{V}_V(f) = \limsup_{\mathcal{P}} \sum_{i=1}^N \sum_{j=1}^M |f(x_{i+1}, y_{j+1}) - f(x_{i+1}, y_j) - f(x_i, y_{j+1}) + f(x_i, y_j)|$$

Note, that the variation of Vitali does not provide a good measure in this case: if the function is constant in  $x$ , for instance, then the variation is zero, regardless how the function changes depending on  $y$ .

## Definition (Hardy-Krause variation)

For 2-dimensional functions the **Hardy-Krause variation** is defined by:

$$\mathcal{V}_{HK}(f) = \mathcal{V}_V(f(x, y)) + \mathcal{V}_V(f(x, 1)) + \mathcal{V}_V(f(1, y))$$



## Theorem (Koksma-Hlawka inequality)

For  $s$ -dimensional functions on  $[0, 1]^s$  the **Koksma-Hlawka inequality** holds

$$\left| \int_{z \in [0, 1]^s} f(z) dz - \frac{1}{N} \sum_{i=1}^N f(z_i) \right| \leq \mathcal{V}_{HK} \cdot \mathcal{D}(z_1, z_2, \dots, z_N),$$

where  $\mathcal{D}(z_1, z_2, \dots, z_N)$  is the discrepancy of the sequence  $z_1, z_2, \dots, z_N$

## Remarks:

- ▶ The Koksma-Hlawka inequality implies that the approximation error by the quasi-Monte Carlo method is  $\mathcal{O}\left(\frac{(\log N)^s}{N}\right)$ , whereas the Monte Carlo method has an error of  $\mathcal{O}\left(\frac{1}{\sqrt{N}}\right)$ .
- ▶ In order to minimize the error there are two orthogonal strategies:
  - ▶ Make function flat by appropriate variable transformations (**importance sampling**)
  - ▶ Use low-discrepancy series (**stratification**)



## Importance sampling in quasi-Monte Carlo integration

Recall that if a function  $f$  needs to be integrated in domain  $V$  and we have a monotonously increasing mapping  $T$  that maps this domain onto  $V'$  we can evaluate the integral over the new domain according to

$$\int_V f(z) dz = \int_{V'} f(T^{-1}(\mathbf{y})) \left| \frac{\partial T^{-1}(\mathbf{y})}{\partial \mathbf{y}} \right| d\mathbf{y},$$

where  $\left| \frac{\partial T^{-1}(\mathbf{y})}{\partial \mathbf{y}} \right|$  is the Jacobi determinant of the inverse transformation.

If quasi-Monte Carlo integration is used, the domain  $V'$  is  $[0, 1]^s$ . Following the arguments used for importance sampling in standard Monte-Carlo integration, it can be shown that the method of importance sampling is independent of whether random samples are transformed, or deterministic samples are transformed.

Note, that this is the basic ingredient of the **Primary Sample Space based MCMC Methods**.



## Summary on quasi-Monte Carlo:

- ▶ May improve speed of convergence
- ▶ Reproducible and simple to parallelize
- ▶ For many functions arising in practice the variation is unbounded
- ▶ For real world settings it is difficult to compute  $\mathcal{D}$  and  $\mathcal{V}_{\mathcal{H}\mathcal{K}}$



Full Siggraph 2019 Course “My favorite Samples”<sup>7</sup> recording ( $\approx 3$  h) available on Youtube.



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<sup>7</sup>A. Keller, I. Georgiev, et al. “My favorite samples”. In: *ACM SIGGRAPH 2019 Courses*. ACM, 2019, pp. 1–271.





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